

6. A class investigated the rate of the reaction between magnesium and hydrochloric acid by measuring the volume of gas produced in 10 seconds using different concentrations of acid.

(a) Tick (✓) the name of the piece of apparatus they used to measure the volume of gas produced. [1]

beaker	☐
thermometer	☐
gas syringe	☐
conical flask	☐

(b) The table shows the results of their experiment.

Concentration of acid (M)	Volume of gas produced (cm ³)			
	Test 1	Test 2	Test 3	Mean
0.2	16	14	15	15
0.4	31	33	30	32
0.6	47	49	29	48
0.8	63	64	65	64
1.0	82	83	79	81

(i) **Circle** the result in the table which was **not** used to calculate a mean.

Give the reason why this result was not used.

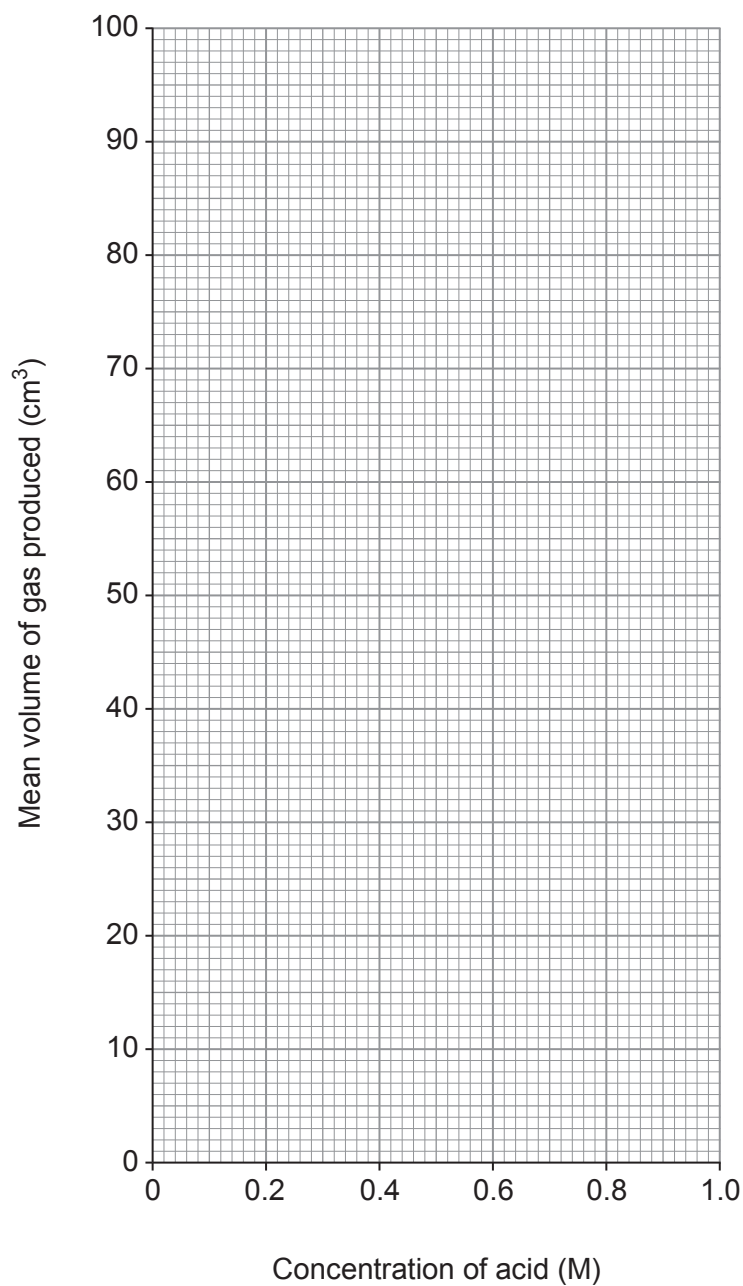
[2]

Reason

.....



- (ii) Plot the mean volume of gas produced against the concentration of acid. Draw a suitable line. [3]



- (iii) Use the information to find the volume of gas produced in 10 seconds using acid of concentration 0.5M. [1]

..... cm³



Examiner
only

(c) In another investigation the class found that as the temperature of the acid increases the reaction is faster.

Tick (✓) the **two** statements which explain why the reaction is faster at a higher temperature.

[2]

- there are more particles of acid☐
- the acid particles are moving faster☐
- there is a higher surface area☐
- there are more collisions per second☐
- the acid particles have less energy☐

(d) The class wanted to identify the gas produced. They tested it as shown in the table.

Test	bubble the gas through limewater	place a burning splint in the gas
Observation	limewater did not change	a squeaky pop and the splint went out

State the name of the gas.

[1]

.....

10

